BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

MATHEMATICS — SSC-I (9th Grade) — ANSWER KEY (Version 2)

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

1. The set of all elements of U which are not in A is called:

Answer: (D) Complement of A 1 mark

The **complement of A**, denoted A^c or A', is defined as $A^c = U - A = \{x \mid x \in U \wedge x \notin A\}$. Difference of two sets A and B is $A - B$. Union combines elements from both; intersection keeps only common elements.

2. If A = {1, 2, 3} and B = {3, 4, 5}, then A ∩ B is:

Answer: (B) {3} 1 mark

The **intersection** $A \cap B$ consists of elements present in **both** sets. Here, the only common element between A and B is 3, so $A \cap B = \{3\}$.

3. If n(A) = 4 and n(B) = 3, then n(A × B) equals:

Answer: (B) 12 1 mark

The number of elements in the Cartesian product is given by $\mathbf{n(A \times B) = n(A) \times n(B)}$. So $n(A \times B) = 4 \times 3 = 12$. Each element of A pairs with each element of B.

4. The set of all first elements of ordered pairs in a binary relation is called:

Answer: (B) Domain 1 mark

In a binary relation R, the set of all **first elements** of the ordered pairs is called the **Domain** of R, and the set of all **second elements** is called the **Range** of R.

5. The factorization of $a^2 - b^2$ is:

Answer: (C) (a + b)(a − b) 1 mark

This is the **difference of two squares** identity: $a^2 - b^2 = (a + b)(a - b)$. $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)^2 = a^2 - 2ab + b^2$ are perfect squares, not differences.

6. The factors of $x^2 + 7x + 12$ are:

Answer: (A) (x + 3)(x + 4) 1 mark

Find two numbers whose **product is 12** and **sum is 7**: these are 3 and 4. So $x^2 + 7x + 12 = x^2 + 3x + 4x + 12 = x(x + 3) + 4(x + 3) = (x + 3)(x + 4)$.

7. The HCF of $12x^2y^3$ and $18xy^2z$ is:

Answer: (B) $6xy^2$ 1 mark

HCF of numerical coefficients 12 and 18 is **6**. The lowest power of x common to both is $\mathbf{x^1}$, and the lowest power of y is $\mathbf{y^2}$. Variable z is absent in the first term, so it is not in the HCF. Thus HCF = $6xy^2$.

8. If the product of HCF and LCM of two polynomials is $x^3 - y^3$, then the product of the polynomials is:

Answer: (A) $x^3 - y^3$ 1 mark

By the key relation: **HCF × LCM = Product of the two polynomials**. So if HCF × LCM = $x^3 - y^3$, then the product $P \times Q = x^3 - y^3$.

9. The solution of 2x + 3 = 11 is:

Answer: (B) 4 1 mark

Subtracting 3 from both sides: $2x = 8$. Dividing by 2: $x = \mathbf{4}$. Check: $2(4) + 3 = 8 + 3 = 11$ √.

10. |−6| equals:

Answer: (B) 6 1 mark

The **absolute value** $|x|$ is the distance of x from 0 on the number line, which is always non-negative. So $|-6| = 6$. By definition: $|x| = x$ if $x \geq 0$, and $|x| = -x$ if $x < 0$.

11. Which of the following is a strict inequality?

Answer: (C) $x > 2$ 1 mark

A **strict inequality** uses only the symbols **<** or **>** (not ≤ or ≥). $x > 2$ uses a strict >. $x \geq 3$ and $x \leq 5$ are non-strict; $x = 4$ is an equation, not an inequality.

12. When both sides of an inequality are multiplied by a negative number, its direction:

Answer: (B) Reverses 1 mark

Multiplying or dividing an inequality by a negative number reverses the direction of the inequality symbol. For example, $-2 < 5$ becomes $(-3)(-2) > (-3)(5)$, i.e., $6 > -15$.

SECTION B — Short Questions (11 × 3 = 33 Marks)

Q.2(i): Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{3, 4, 5, 6, 7\}$. Find (a) $A \cup B$ (b) $A \cap B$ (c) $A - B$ 3 marks

(a) **Union** $A \cup B = \{x \mid x \in A \lor x \in B\} = \mathbf{\{1, 2, 3, 4, 5, 6, 7\}}$ 1 mark

(b) **Intersection** $A \cap B = \{x \mid x \in A \land x \in B\} = \mathbf{\{3, 4, 5\}}$ 1 mark

(c) **Difference** $A - B = \{x \mid x \in A \land x \notin B\} = \mathbf{\{1, 2\}}$ 1 mark

OR

Q.2(i) OR: Let $U = \{1, 2, 3, \ldots, 10\}$ and $A = \{2, 4, 6, 8\}$. Find (a) A^c (b) $n(A)$ (c) $n(U)$ 3 marks

(a) **Complement** $A^c = U - A = \{x \mid x \in U \land x \notin A\} = \mathbf{\{1, 3, 5, 7, 9, 10\}}$ 1 mark

(b) $n(A)$ = number of elements in A = **4** 1 mark

(c) $n(U)$ = number of elements in U = **10** 1 mark

Q.2(ii): Write each set in set-builder form: (a) $\{2, 4, 6, 8, 10\}$ (b) $\{1, 4, 9, 16, 25\}$ (c) $\{-3, -2, -1, 0, 1, 2, 3\}$ 3 marks

In set-builder form a set is written as $\{x \mid x \text{ satisfies some rule}\}$.

(a) $\{2, 4, 6, 8, 10\} = \mathbf{\{2n \mid n \in N \land 1 \leq n \leq 5\}}$ (even natural numbers up to 10) 1 mark

(b) $\{1, 4, 9, 16, 25\} = \mathbf{\{n^2 \mid n \in N \land 1 \leq n \leq 5\}}$ (squares of first five natural numbers) 1 mark

(c) $\{-3, -2, -1, 0, 1, 2, 3\} = \mathbf{\{x \mid x \in Z \land -3 \leq x \leq 3\}}$ 1 mark

OR

Q.2(ii) OR: If $A = \{1, 2, 3\}$ and $B = \{3, 4\}$, find $A \times B$ and $B \times A$. Show that $A \times B \neq B \times A$. 3 marks

The Cartesian product $A \times B$ is the set of all ordered pairs (a, b) with $a \in A$ and $b \in B$.

$A \times B = \mathbf{\{(1, 3), (1, 4), (2, 3), (2, 4), (3, 3), (3, 4)\}}$ 1 mark

$B \times A = \mathbf{\{(3, 1), (3, 2), (3, 3), (4, 1), (4, 2), (4, 3)\}}$ 1 mark

For example, $(1, 3) \in A \times B$ but $(1, 3) \notin B \times A$. Therefore, $\mathbf{A \times B \neq B \times A}$. 1 mark

Q.2(iii): In a class of 50 students, 30 like Mathematics, 25 like Physics, and 10 like both. How many students like at least one subject? How many like neither? 3 marks

Let M = set of students who like Mathematics, P = set of students who like Physics.

$n(U) = 50, \quad n(M) = 30, \quad n(P) = 25, \quad n(M \cap P) = 10$ ½ mark

Using the formula $\mathbf{n(M \cup P) = n(M) + n(P) - n(M \cap P)}$:

$n(M \cup P) = 30 + 25 - 10 = \mathbf{45}$ students like at least one subject. 1½ marks

Students who like neither = $n(U) - n(M \cup P) = 50 - 45 =$ **5 students**. 1 mark

OR

Q.2(iii) OR: Find the values of x and y if $(2x - 1, y + 3) = (5, 7)$. 3 marks

By the **equality of ordered pairs**, $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.

So: $2x - 1 = 5$ ...(i) and $y + 3 = 7$ ...(ii) 1 mark

From (i): $2x = 6 \Rightarrow \mathbf{x = 3}$ 1 mark

From (ii): $\mathbf{y = 4}$ 1 mark

Q.2(iv): Let R = {(1, 2), (2, 4), (3, 6), (4, 8)}. Find (a) Domain of R (b) Range of R (c) R^{-1} 3 marks

(a) **Domain of R** = set of all first elements of the ordered pairs = **{1, 2, 3, 4}** 1 mark

(b) **Range of R** = set of all second elements of the ordered pairs = **{2, 4, 6, 8}** 1 mark

(c) **Inverse R^{-1}** is obtained by swapping each ordered pair (a, b) with (b, a):

R^{-1} = **{(2, 1), (4, 2), (6, 3), (8, 4)}** 1 mark

OR

Q.2(iv) OR: If A = {a, b, c}, find the number of all possible binary relations in A × A. 3 marks

The number of binary relations on a set is $2^{n(A \times A)}$.

First, find $n(A \times A)$: 1 mark

$n(A) = 3$, so $n(A \times A) = n(A) \times n(A) = 3 \times 3 = 9$

Number of binary relations in $A \times A = 2^{n(A \times A)} = \mathbf{2^9 = 512}$ 2 marks

Q.2(v): Factorize: (a) $x^2 - 9x + 20$ (b) $4x^2 - 25y^2$ 3 marks

(a) $x^2 - 9x + 20$: Find two numbers whose product is +20 and sum is −9. These are −4 and −5. 1½ marks

$x^2 - 9x + 20 = x^2 - 4x - 5x + 20$

$= x(x - 4) - 5(x - 4)$

$= \mathbf{(x - 4)(x - 5)}$

(b) $4x^2 - 25y^2$: Apply the difference of squares identity $a^2 - b^2 = (a + b)(a - b)$. 1½ marks

$4x^2 - 25y^2 = (2x)^2 - (5y)^2$

$= \mathbf{(2x + 5y)(2x - 5y)}$

OR

Q.2(v) OR: Factorize: (a) $x^3 + 27$ (b) $8a^3 - 27b^3$ 3 marks

Use the identities $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ and $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$.

(a) $x^3 + 27 = x^3 + 3^3 = \mathbf{(x + 3)(x^2 - 3x + 9)}$ 1½ marks

(b) $8a^3 - 27b^3 = (2a)^3 - (3b)^3$ 1½ marks

$= \mathbf{(2a - 3b)(4a^2 + 6ab + 9b^2)}$

Q.2(vi): Factorize the trinomial: $3x^2 + 11x + 6$ 3 marks

Product of extreme coefficients = $3 \times 6 = 18$. Find two numbers with product 18 and sum 11: **9 and 2**. 1 mark

$3x^2 + 11x + 6 = 3x^2 + 9x + 2x + 6$

$= 3x(x + 3) + 2(x + 3)$ 1 mark

$= \mathbf{(x + 3)(3x + 2)}$ 1 mark

OR

Q.2(vi) OR: Factorize: $x^4 + x^2y^2 + y^4$ 3 marks

Add and subtract x^2y^2 to make the first and last terms form a perfect square: 1 mark

$x^4 + x^2y^2 + y^4 = x^4 + 2x^2y^2 + y^4 - x^2y^2$

$= (x^2 + y^2)^2 - (xy)^2$ 1 mark

Using the difference of two squares $a^2 - b^2 = (a + b)(a - b)$:

$= \mathbf{(x^2 + y^2 + xy)(x^2 + y^2 - xy)}$ 1 mark

Q.2(vii): Find the HCF of $x^2 - 4$ and $x^2 + 4x + 4$ 3 marks

Factorize each polynomial completely:

$x^2 - 4 = (x + 2)(x - 2)$ (difference of squares) 1 mark

$x^2 + 4x + 4 = (x + 2)^2$ (perfect square) 1 mark

The common factor with the least power is $(x + 2)$. Therefore:

HCF = $(x + 2)$ 1 mark

OR

Q.2(vii) OR: Find the LCM of the monomials: $6x^2y$, $9xy^2$, $12x^3$ 3 marks

Find the LCM of the numerical coefficients 6, 9, 12:

$6 = 2 \cdot 3, \quad 9 = 3^2, \quad 12 = 2^2 \cdot 3$

LCM of coefficients = $2^2 \cdot 3^2 = \mathbf{36}$ 1 mark

For each variable, take the highest power that appears:

- x: highest power is x^3
- y: highest power is y^2 1 mark

Therefore **LCM = $36x^3y^2$**. 1 mark

Q.2(viii): Simplify: $(x^2 - 1)/(x^2 + 2x + 1) \times (x + 1)/(x - 1)$ 3 marks

Factorize numerators and denominators completely:

$(x^2 - 1) = (x + 1)(x - 1)$

$(x^2 + 2x + 1) = (x + 1)^2$ 1 mark

Substitute back:

$= [(x + 1)(x - 1) / (x + 1)^2] \times [(x + 1) / (x - 1)]$ 1 mark

Cancel common factors:

$= (x - 1) / (x + 1) \times (x + 1) / (x - 1)$

$=$ **1** 1 mark

OR

Q.2(viii) OR: Find the square root of $4x^2 + 12xy + 9y^2$ by factorization. 3 marks

Write the given expression as a perfect square by recognizing the pattern $a^2 + 2ab + b^2 = (a + b)^2$:

$4x^2 + 12xy + 9y^2 = (2x)^2 + 2(2x)(3y) + (3y)^2$ 1 mark

$= (2x + 3y)^2$ 1 mark

Applying square root on both sides:

$\sqrt{(4x^2 + 12xy + 9y^2)} = \pm(2x + 3y)$ 1 mark

Q.2(ix): Solve for x: $(3x - 1)/4 = (x + 2)/3$ 3 marks

Cross multiply: 1 mark

$3(3x - 1) = 4(x + 2)$

$9x - 3 = 4x + 8$

Move variables to one side and constants to the other: 1 mark

$9x - 4x = 8 + 3$

$5x = 11$

$x =$ **11/5** 1 mark

OR

Q.2(ix) OR: Solve the radical equation: $\sqrt{(x + 5)} = 4$. Check your solution. 3 marks

Square both sides to eliminate the radical: 1 mark

$(\sqrt{(x + 5)})^2 = 4^2$

$x + 5 = 16$

$x =$ **11** 1 mark

Check: substitute $x = 11$ in the original equation:

$\sqrt{(11 + 5)} = \sqrt{16} = 4$ √ Solution verified. 1 mark

Q.2(x): Solve: $|2x - 3| = 5$ **3 marks**

An equation of the form $|ax + b| = c$ (where $c > 0$) is equivalent to two equations: $ax + b = c$ or $ax + b = -c$. **1 mark**

Case 1: $2x - 3 = 5 \Rightarrow 2x = 8 \Rightarrow \mathbf{x = 4}$

Case 2: $2x - 3 = -5 \Rightarrow 2x = -2 \Rightarrow \mathbf{x = -1}$ **1 mark**

Therefore, the solution set is $\mathbf{\{-1, 4\}}$. **1 mark**

OR

Q.2(x) OR: Solve the absolute value equation: $|x - 4| = 7$ **3 marks**

Remove the absolute value bars to get two equations: **1 mark**

Case 1: $x - 4 = 7 \Rightarrow \mathbf{x = 11}$

Case 2: $x - 4 = -7 \Rightarrow \mathbf{x = -3}$ **1 mark**

The solution set is $\mathbf{\{-3, 11\}}$. **1 mark**

Q.2(xi): Solve the inequality: $3x - 5 < 2x + 7, x \in R$ **3 marks**

Subtract 2x from both sides: **1 mark**

$3x - 2x - 5 < 7$

$x - 5 < 7$

Add 5 to both sides: **1 mark**

$x < 12$

Solution set: $\mathbf{\{x \mid x \in R \wedge x < 12\}}$ **1 mark**

OR

Q.2(xi) OR: Solve the compound inequality: $-3 < 2x + 1 \leq 7, x \in R$ **3 marks**

Apply operations to all three parts simultaneously.

Subtract 1 from each part: **1 mark**

$-3 - 1 < 2x + 1 - 1 \leq 7 - 1$

$-4 < 2x \leq 6$

Divide each part by 2 (positive, so direction unchanged): **1 mark**

$-2 < x \leq 3$

Solution set: $\mathbf{\{x \mid x \in R \wedge -2 < x \leq 3\}}$ **1 mark**

SECTION C — Long Questions (4 × 5 = 20 Marks)

Q.3(i): Verify the associative law of union (A ∪ B) ∪ C = A ∪ (B ∪ C) both analytically and by a Venn diagram. U = {1, …, 9}, A = {1, 2, 3}, B = {3, 4, 5}, C = {5, 6, 7} 5 marks

Analytical verification: 3 marks

LHS = (A ∪ B) ∪ C

A ∪ B = {1, 2, 3} ∪ {3, 4, 5} = {1, 2, 3, 4, 5}

(A ∪ B) ∪ C = {1, 2, 3, 4, 5} ∪ {5, 6, 7} = **{1, 2, 3, 4, 5, 6, 7}** 1½ marks

RHS = A ∪ (B ∪ C)

B ∪ C = {3, 4, 5} ∪ {5, 6, 7} = {3, 4, 5, 6, 7}

A ∪ (B ∪ C) = {1, 2, 3} ∪ {3, 4, 5, 6, 7} = **{1, 2, 3, 4, 5, 6, 7}** 1½ marks

Since LHS = RHS, the **associative law of union is verified**.

Venn diagram verification: 2 marks

```
U = {1,2,3,4,5,6,7,8,9}

LHS: (A U B) U C            RHS: A U (B U C)
+---------------+           +---------------+
|      ///      |           |      ///      |
|  +-----+-+-----+           |  +-----+-+-----+
|  | ///1|3|4//5 |           |  | ///1|3|4//5 |
|  | A///|/|//B  |           |  | A///|/|//B  |
|  |  2//|/|///  |           |  |  2//|/|///  |
|  +-----+-+--+  |           |  +-----+-+--+  |
|          | C|  |           |          | C|  |
|          |5/|  |           |          |5/|  |
|          |6/|  |           |          |6/|  |
|          |7/|  |           |          |7/|  |
|  8  9    +--+  |           |  8  9    +--+  |
+---------------+           +---------------+
```

Both diagrams shade the same region {1, 2, 3, 4, 5, 6, 7}, confirming the associative law.

OR

Q.3(i) OR: Verify the distributive law A ∪ (B ∩ C) = (A ∪ B) ∩ (A ∪ C) for A = {1, 2, 3, 4}, B = {3, 4, 5, 6}, C = {5, 6, 7, 8} 5 marks

LHS = A ∪ (B ∩ C) 2 marks

B ∩ C = {3, 4, 5, 6} ∩ {5, 6, 7, 8} = {5, 6}

A ∪ (B ∩ C) = {1, 2, 3, 4} ∪ {5, 6} = **{1, 2, 3, 4, 5, 6}**

RHS = (A ∪ B) ∩ (A ∪ C) 2 marks

A ∪ B = {1, 2, 3, 4} ∪ {3, 4, 5, 6} = {1, 2, 3, 4, 5, 6}

A ∪ C = {1, 2, 3, 4} ∪ {5, 6, 7, 8} = {1, 2, 3, 4, 5, 6, 7, 8}

(A ∪ B) ∩ (A ∪ C) = {1, 2, 3, 4, 5, 6} ∩ {1, 2, 3, 4, 5, 6, 7, 8} = **{1, 2, 3, 4, 5, 6}**

Since LHS = RHS = {1, 2, 3, 4, 5, 6}, the **distributive law of union over intersection is verified**. 1 mark

Q.3(ii): Factorize completely: (a) $x^4 + 4y^4$ (b) $(x^2 - 5x + 2)(x^2 - 5x + 4) - 3$ **5 marks**

(a) $x^4 + 4y^4$: **2½ marks**

Add and subtract $4x^2y^2$ to complete the square:

$x^4 + 4y^4 = x^4 + 4x^2y^2 + 4y^4 - 4x^2y^2$

$= (x^2 + 2y^2)^2 - (2xy)^2$

Using $a^2 - b^2 = (a + b)(a - b)$:

$= (x^2 + 2y^2 + 2xy)(x^2 + 2y^2 - 2xy)$

(b) $(x^2 - 5x + 2)(x^2 - 5x + 4) - 3$: **2½ marks**

Let $y = x^2 - 5x$. Then the expression becomes:

$(y + 2)(y + 4) - 3$

$= y^2 + 6y + 8 - 3$

$= y^2 + 6y + 5$

$= (y + 1)(y + 5)$

Replace y with $x^2 - 5x$:

$= (x^2 - 5x + 1)(x^2 - 5x + 5)$

OR

Q.3(ii) OR: Find HCF and LCM of $x^3 + 2x^2 - 3x$ and $x^3 + x^2 - 6x$. Verify that HCF × LCM = product of the polynomials. **5 marks**

Factorize each polynomial completely: **2 marks**

$P = x^3 + 2x^2 - 3x = x(x^2 + 2x - 3) = x(x + 3)(x - 1)$

$Q = x^3 + x^2 - 6x = x(x^2 + x - 6) = x(x + 3)(x - 2)$

Common factors with lowest powers give the HCF: **1 mark**

HCF = $x(x + 3)$

All factors with highest powers give the LCM: **1 mark**

LCM = $x(x + 3)(x - 1)(x - 2)$

Verification: **1 mark**

$\text{HCF} \times \text{LCM} = x(x + 3) \cdot x(x + 3)(x - 1)(x - 2) = x^2(x + 3)^2(x - 1)(x - 2)$

$P \times Q = x(x + 3)(x - 1) \cdot x(x + 3)(x - 2) = x^2(x + 3)^2(x - 1)(x - 2)$

Therefore, **HCF × LCM = P × Q**. Verified.

Q.3(iii): Find the square root of $x^4 - 6x^3 + 13x^2 - 12x + 4$ by the division method and state whether it is a perfect square. 5 marks

Applying the long-division method for square roots of polynomials:

```
                  x^2  - 3x  + 2         <- quotient (square root)
              _______________________
         x^2 | x^4 - 6x^3 + 13x^2 - 12x + 4
              x^4
              ------
  2x^2 - 3x |      - 6x^3 + 13x^2 - 12x + 4
                   - 6x^3 +  9x^2
                   ---------------
2x^2 - 6x + 2 |             4x^2 - 12x + 4
                            4x^2 - 12x + 4
                            ---------------
                                    0      <- remainder
```

Step-by-step: 3 marks

1. First term of square root: $\sqrt{(x^4)} = x^2$. Subtract x^4 from dividend. Remainder: $-6x^3 + 13x^2 - 12x + 4$.
2. Double the current root: $2x^2$. Divide $-6x^3$ by $2x^2$ to get $-3x$. Add to divisor: $2x^2 - 3x$. Multiply $(2x^2 - 3x)(-3x) = -6x^3 + 9x^2$. Subtract: $4x^2 - 12x + 4$.
3. Double the current root: $2x^2 - 6x$. Divide $4x^2$ by $2x^2$ to get $+2$. Add to divisor: $2x^2 - 6x + 2$. Multiply by 2: $4x^2 - 12x + 4$. Subtract: remainder = 0.

Therefore, $\mathbf{\sqrt{(x^4 - 6x^3 + 13x^2 - 12x + 4)} = \pm(x^2 - 3x + 2)}$ 1 mark

Since the remainder is 0, the given expression **IS a perfect square**. 1 mark

OR

Q.3(iii) OR: Simplify: $((x^2 - 4)/(x^2 - 9)) \div ((x + 2)/(x + 3)) \times ((x^2 - 6x + 9)/(x^2 + 2x))$ 5 marks

Factorize every numerator and denominator: 2 marks

$x^2 - 4 = (x + 2)(x - 2)$

$x^2 - 9 = (x + 3)(x - 3)$

$x^2 - 6x + 9 = (x - 3)^2$

$x^2 + 2x = x(x + 2)$

Rewrite, converting division into multiplication by the reciprocal: 1 mark

$= [(x + 2)(x - 2) / ((x + 3)(x - 3))] \times [(x + 3)/(x + 2)] \times [(x - 3)^2 / (x(x + 2))]$

Multiply numerators and denominators together: 1 mark

$= [(x + 2)(x - 2)(x + 3)(x - 3)^2] / [(x + 3)(x - 3)(x + 2)(x)(x + 2)]$

Cancel common factors $(x + 2)$, $(x + 3)$, and one $(x - 3)$:

$\mathbf{= [(x - 2)(x - 3)] / [x(x + 2)]}$ 1 mark

Q.3(iv): Solve the compound inequality $5 < 2x + 3 \leq 17$, $x \in R$. Show the solution set on a number line. 5 marks

A three-part compound inequality is solved by applying the same operation to all three parts.

Subtract 3 from each part: 1½ marks

$5 - 3 < 2x + 3 - 3 \leq 17 - 3$

$2 < 2x \leq 14$

Divide each part by 2 (positive, direction preserved): 1½ marks

$1 < x \leq 7$

Solution set: $\{x \mid x \in R \wedge 1 < x \leq 7\}$ 1 mark

Number line representation: hollow circle at 1 (excluded), filled dot at 7 (included), shaded segment between. 1 mark

```
<---|---|---|---|---|---|---|---|---|---|--->
    -1   0   1   2   3   4   5   6   7   8
             ○===================●
             1                     7
(Hollow circle at 1 means 1 is excluded; filled dot at 7 means 7 is included.)
```

OR

Q.3(iv) OR: The sum of three consecutive integers is greater than 30 and less than or equal to 72. Find all possible values of the smallest integer. 5 marks

Let the three consecutive integers be n, n + 1, and n + 2, where n is the smallest.

Their sum is: 1 mark

$n + (n + 1) + (n + 2) = 3n + 3$

Given the compound inequality:

$30 < 3n + 3 \leq 72$ 1 mark

Subtract 3 from each part: 1 mark

$27 < 3n \leq 69$

Divide each part by 3: 1 mark

$9 < n \leq 23$

Since n must be an integer, the smallest integer n can take values:

$n \in \{10, 11, 12, 13, \ldots, 23\}$ (14 possible values) 1 mark

For example, the smallest triple is (10, 11, 12) with sum 33, and the largest is (23, 24, 25) with sum 72.